Status	Finished
Started	Tuesday, 20 October 2026, 5:33 PM
Completed	Tuesday, 20 October 2026, 5:37 PM
Duration	4 mins 3 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Merge Sort
- b. Linear Search
- c. Binary Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Quick sort
- b. Merge sort
- c. Heap sort
- d. Selection sort

The correct answer is:
Merge sort

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting is rarely used in programming
- b. Sorting helps in finding duplicates quickly
- c. Sorting makes it harder to search for items
- d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **4**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By using the bubble-up method
- b. By sorting data in a single pass
- c. By breaking the input into smaller parts and merging them
- d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **5**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. Duplicates
- c. Selection
- d. All of the above

The correct answer is: All of the above

Question **6**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Dynamic programming
- b. Greedy method
- c. Backtracking
- d. Divide and conquer

The correct answer is:
Divide and conquer

Question **7**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Searching
- b. Complexity
- c. Sorting
- d. Merging

The correct answer is:
Complexity

Question **8**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Arranging data in a particular format
- c. Deleting data from a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **9**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the smallest element
- b. Bubbling up the largest element to its correct position
- c. Dividing the list into halves
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It simplifies the sorting process by using only one pass
- b. It avoids the need for recursion
- c. It only works on small datasets
- d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **11**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Linear Search
- b. Binary Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **12**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It makes the search operation slower
- b. It allows for faster searching
- c. It increases the number of comparisons needed
- d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **13**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort is less efficient for large lists
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Provides a basis for other algorithms to work efficiently
b. Decreases the efficiency of algorithms
c. Makes data harder to manage
d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **15**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It is based on the divide-and-conquer approach
c. It does not require recursion
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach